

Guillaume Bogard



I am a senior software engineer with a strong focus on statically-typed languages and functional programming. I've been programming for about 16 years, and professionally so for more than 10 years, on both the back and front ends.

Nowadays, I mostly use Haskell and Typescript professionally, but my experience spans multiple languages including 4 years of commercial Scala, and 2 years of commercial experience with Rust. My achievements include building AI agents and RAG systems for Aletiq, and building a web-scale crawler for Qwant.

📍 Paris, FR

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Work experience

Senior Software Engineer (Haskell, TypeScript, React)

April 2025 – Present

Aletiq · [haskell](#) · [react](#) · [nix](#)

- Building a RAG-based AI assistant that lets users query technical documents (ISO standards, assembly drawings, industrial specs) in natural language
- Building a large-scale PDF indexer that ingests thousands of customer documents into the RAG system
- Shipping AI-powered suggestions for related documents and products, improving traceability across the PLM
- Full-stack development in TypeScript (Effect.ts, React) and Haskell, alongside ongoing LLM research

Rust Software Engineer @Qwant (Crawler Team)

December 2022 – April 2025

Qwant · [rust](#) · [kubernetes](#) · [python](#)

- Built a high-performance HTML content extraction service in Rust, matching industry-standard quality (e.g. Trafilatura) while running orders of magnitude faster
- Architected a sharded, lock-free crawler that dramatically improved throughput while enforcing strict politeness policies (no more than one concurrent download per website)
- Built a service that downloads, caches and parses robots.txt files, then filters URLs against the rules they contain
- Worked across the crawler's core: URL redirection, URL filtering and end-to-end testing

Lead Scala developer @Canal+

January 2021 – December 2022

Canal+ · [kafka](#) · [kubernetes](#) · [scala](#)

- Led the documentation services team on Mediasub, a Scala/Typelevel media asset management (MAM) platform ingesting 300+ hours of content per day
- Architected core services for video annotation, faceted full-text search and evolving concept metadata
- Drove the migration of millions of legacy XML documents through a high-throughput Kafka/fs2 pipeline, enabling legacy systems to be shut down
- Mentored the team in functional programming: Cats Effect, hexagonal architecture and property-based testing (ScalaCheck)

Scala developer @Canal+

October 2019 – January 2021

Linkvalue · Canal+ · [kafka](#) · [scala](#)

- Built Scala/Typelevel services for Mediasub, a next-generation media asset management platform
- Developed annotation tooling for marking up precise moments in videos with participant, action and topic metadata
- Built faceted full-text search that let journalists find exact moments across millions of videos
- Migrated millions of legacy XML documents while preserving historical annotations

Functional programming tutor @HEI, Graduate School of Engineering

October 2019 – October 2019

École des Hautes Études d'Ingénieur (HEI) · [scala](#)

- Taught a 15-hour functional programming course in Scala to engineering students (referential transparency, totality, laziness, composition, IO monads and functional streams with Cats Effect and fs2)
- Guided students through a project: a purely functional Tic-Tac-Toe played over TCP

[Linkvalue](#) · [Colisweb](#) · [scala](#) · [kubernetes](#)

- Developed Scala microservices in a purely functional style (Cats, Http4s, Tapir)
- Migrated services from Play Framework to the Typelevel stack, unifying libraries and coding style across the codebase
- Built an open-source tracing library (scala-opentracing) that improved the platform's observability
- Deployed services to Kubernetes and mentored an intern in functional programming

Full-stack Javascript developer @Decathlon Maps

January 2019 – May 2019

[Linkvalue](#) · [Decathlon](#) · [react](#) · [node](#) · [javascript](#)

- Built front-end and back-end features for a sports-places PWA: React/Redux Saga, Node.js/Koa and PostgreSQL
- Introduced end-to-end tests (Cypress), code reviews and pre-commit linting, replacing a push-to-master workflow
- Refactored the codebase and reduced its size by around 30%
- Taught functional programming principles (composition, referential transparency) and introduced Redux Saga to separate business logic from side effects

Front-end developer @Norauto

June 2018 – January 2019

[Linkvalue](#) · [Norauto](#) · [react](#) · [vue](#) · [polymer](#) · [javascript](#)

- Built progressive web apps (React, Vue) as part of a full overhaul of Norauto's information system
- Co-built a framework-agnostic design system with custom elements (Polymer), documented and shared company-wide
- Applied a micro-frontend architecture, splitting the domain into independent, loosely coupled applications
- Introduced styled-components, React Context and state-management practices to the team

Front-end developer @Qlikbeton

January 2018 – June 2018

[Linkvalue](#) · [Qlikbeton](#) · [react](#) · [javascript](#)

- Built the front end of a multi-country concrete marketplace in React, in a team of four
- Designed a per-country configuration system to override styling and behaviour without duplicating code
- Kept the codebase functional (Ramda, immutable data, isolated side effects) with 100% unit test coverage

Full-Stack developer @Pingflow

November 2016 – December 2017

[Pingflow](#) · [vue](#) · [node](#) · [scala](#) · [javascript](#) · [kubernetes](#)

- Led a small team and rewrote Pingview, the company's dashboard SaaS, into a stateless, horizontally scalable architecture
- Migrated the stack from vanilla JavaScript/jQuery to TypeScript and Vue.js
- Introduced Redis caching and pub/sub for live updates (thousands per second), and Scala/Akka to model data flows as actors and messages
- Shipped the app as Docker images on a Rancher cluster

Junior Javascript Developer @Docler Holding

May 2016 – November 2016

[Docler Holding](#) · [javascript](#) · [node](#)

- Built a SPA (AngularJS) that let marketers manage e-mail campaigns end to end, including a WYSIWYG e-mail designer
- Built an e-mail tracking system measuring opens, click rates and per-recipient interaction heatmaps
- Contributed to the Node.js/StrongLoop back end

Skills

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- **Languages:** Strong knowledge of Haskell, Scala, Rust and TypeScript with some notions of Java
 - **Libraries and frameworks:** Strong knowledge of the Typelevel stack (Cats, Cats Effect...), significant experience with Effect (TypeScript)
 - **Infrastructure & data:** Nix, Postgres, Kafka
 - **Frontend:** React with Tanstack Query, Vue
 - Significant experience with agentic workflows